

Ali Mohajeri

Electrical Engineer | Embedded Systems, Motor Drives, and Robotics

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Portfolio | LinkedIn

Professional Summary

Electrical engineer working at the intersection of power electronics, embedded systems, and control. Experienced in developing complete electronic systems from architecture and PCB design to firmware, integration, and testing, with applications in robotics and industrial automation.

Technical Skills

Embedded Systems	STM32, C/C++, real-time control, CAN, SPI, UART, I ² C, RS-485/422, Raspberry Pi integration
Power Electronics	PMSM/BLDC motor drives, inverter design, gate drivers, current and voltage sensing, precharge circuits, protection systems, DC-DC converters
Control	Field-oriented control, current/velocity/position loops, impedance control, torque control, PID, state-space control, Gaussian-process modelling
Hardware and Tools	Altium Designer, multilayer PCB design, MATLAB/Simulink, Python, COMSOL, PSIM, NI Multisim, Git, L ^A T _E X

Professional Experience

8Robotics <i>Electrical Engineer</i>	Jul. 2025 – Present <i>Tehran, Iran</i>
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- Designed a 2 kW PMSM drive for humanoid robot joint actuators, covering the power stage, sensing, protection circuitry, 4-layer PCB, embedded firmware, and control implementation.
- Integrated an absolute magnetic encoder and implemented precise position, impedance, velocity, and torque control for robotic joint operation.
- Developed CAN-based command and feedback communication together with overcurrent, overvoltage, and stall/locked-rotor shutdown logic.
- Architected a dual-STM32F7 interface board that connects a Raspberry Pi master to distributed joint drives through eight independent CAN buses while supporting a 1 kHz real-time command cycle.
- Designed a 48 V power distribution board with independently controlled actuator clusters, controlled precharge, high-side MOSFET switching, current sensing, voltage monitoring, isolated gate drive, and regenerative-energy protection.

Javaneh Narmafzar <i>Electrical Engineer</i>	Jan. 2024 – Mar. 2025 <i>Tehran, Iran</i>
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- Contributed to the design and development of a 27 kW axial-flux PMSM drive for a semi-autonomous industrial vehicle prototype.
- Designed and evaluated the IGBT-based inverter power stage, gate-drive circuitry, DC-link architecture, current sensing, and hardware protection functions.
- Implemented field-oriented control on STM32, including current regulation, velocity control, fault handling, and motor-drive bring-up and testing.

Education

University of Tehran <i>M.Sc. in Electrical Engineering, Control</i>	Sep. 2024 – Present <i>Tehran, Iran</i>
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- GPA: 18.26/20

University of Tehran <i>B.Sc. in Electrical Engineering, Control</i>	Sep. 2020 – Sep. 2024 <i>Tehran, Iran</i>
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- GPA: 18.50/20; ranked 4th in the Control track and 12th among 120 Electrical Engineering students.
- Thesis: *Design and Implementation of a Digital Impedance-Spectroscopy System.*

Languages

Persian: Native | English: Fluent — IELTS 7.0 (Listening 7.5, Reading 8.0, Writing 6.0, Speaking 6.5)

Selected Research Project

Gaussian-Process-Based Feedback Linearization

Control and Machine Learning

- Developed a Gaussian-process model to estimate uncertain nonlinear dynamics and incorporated it into a feedback-linearization controller to improve trajectory tracking under modelling uncertainty.

Research Experience

MEMS Laboratory, University of Tehran

Research Assistant

Jul. 2023 – Jul. 2025

Tehran, Iran

- Designed a digital impedance-spectroscopy system for biological sample characterization, including a low-noise analog front end, signal conditioning, STM32-based data acquisition, and PCB implementation.
- Co-authored an IEEE conference paper on simulation-based analysis in COMSOL

Additional Academic Experience

- Granted direct admission to the M.Sc. program at the University of Tehran based on academic performance.
- Teaching Assistant for courses including Linear Control Systems, Electrical Machines, Engineering Mathematics, Probability and Statistics, Communication Systems, Instrumentation and Control, Industrial Control, and Biomedical Engineering.
- Ranked 107th among approximately 155,000 participants in Iran's national university entrance examination in Mathematics and Physics.